

Name: _____

Teacher/Class: _____

SYDNEY TECHNICAL HIGH SCHOOL

HSC ASSESSMENT TASK 1

DECEMBER 2005

MATHEMATICS

Time Allowed: 70 minutes

Instructions:

- Start each question on a new page.
- You may write on the front and back of each sheet of paper. Ask for more paper if required.
- Indicated marks are a guide only and may be changed slightly if necessary.
- Marks may not be awarded for careless or badly arranged work.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	Total
/7	/7	/7	/7	/8	/8	/8	/9	/61

QUESTION 1:**Marks**

Differentiate:

i) $2x^3 + x + 4$

1

ii) $(5x^2 - 3)^4$

2

iii) $\frac{3x}{2x+5}$

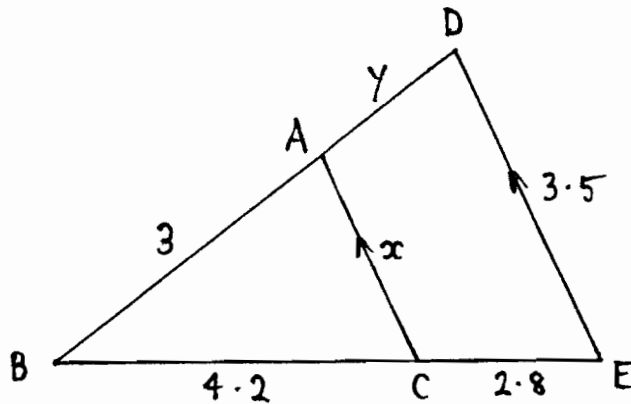
2

iv) $(x-3)(x^3 + 5x - 6)$

2**QUESTION 2:** (Start new page)A. Show that the point $(2, -8)$ lies on the curve $y = 3x^3 - 8x^2$ **1**

i) Hence find the equation of the tangent at this point

3B. Find the equation of the locus of the point $P(x, y)$ if $PA = PB$ where A is the point $(-2, 1)$ and B is $(4, -3)$ **3**

QUESTION 3: (Start new page)**Marks****A.**

- i) By which test is $\triangle ABC \parallel \triangle DBE$? 1
(DO NOT PROVE)

- ii) Hence or otherwise find the value of x and y by giving a reason and showing all working 3

- B.** i) Another way of writing $\sqrt{2x-4}$ is $(2x-4)^a$. Find a 1

- ii) Hence differentiate $x^2 \sqrt{2x-4}$ 2

QUESTION 4: (Start a new page)

- A.** Let the roots of $x^2 + 3x - 5 = 0$ be α and β . Without solving the equation, find the values of: 4

i) $\alpha + \beta$ ii) $\alpha \beta$

iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ iv) $\alpha^2 + \beta^2$

- B.** By making a suitable substitution, solve $3^{2x} - 10 \cdot 3^x + 9 = 0$ 3

QUESTION 5: (Start a new page)

Marks

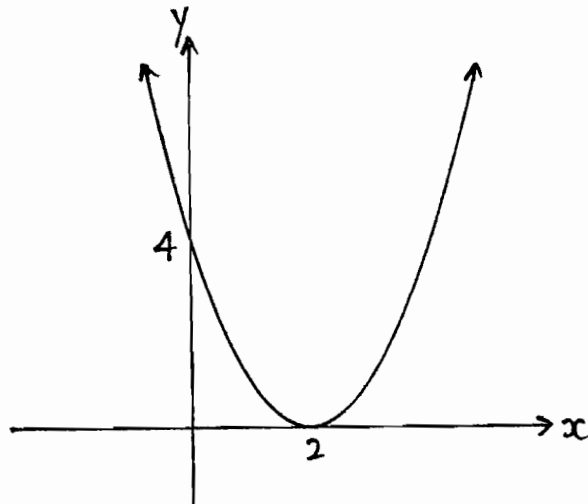
A. A parabola has an equation $y = x^2 - 6x - 7$

i) Sketch this parabola showing its vertex and the x and y intercepts 4

ii) Hence or otherwise solve $x^2 - 6x - 7 > 0$ 1

iii) What is the minimum value of this parabola? 1

B. This diagram could be the graph of: 1



(A) $y = x^2 + 4$

(B) $y = x^2 - 2x$

(C) $y = x^2 + 4x$

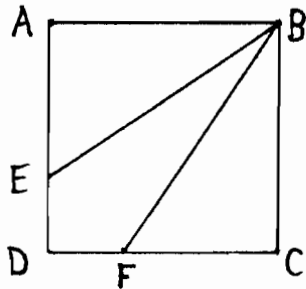
(D) $y = (x - 2)^2$

(E) $y = (x + 2)^2$

C. Sketch a negative definite quadratic function. 1

QUESTION 6: (Start a new page)**Marks**

- A. Find the equation of the locus of the point $P(x, y)$ which moves so that it is always a distance of 3 units from the point $(2, -1)$. 2

B.

ABCD is a square

ED = FD

- i) Prove that triangles BAE and BCF are congruent. 3
- ii) Given $\angle BFC = 55^\circ$, find $\angle EBF$ giving reasons 3

QUESTION 7: (Start a new page)

- A. The vertex of a parabola is $(1, 4)$ and its directrix is $x = -3$
- i) Sketch the parabola 1
- ii) Find the focal length “a” of this parabola and hence give the coordinates of the focus. 2
- iii) Find the equation of this parabola 2
- B. Find the values of a, b and c if 3
- $$2x^2 + 3x - 5 \equiv a(x+1)^2 + b(x+1) + c$$

QUESTION 8: (Start a new page)

Marks

- | | | |
|-----------|--|----------|
| A. | Find the range of values of k for which the roots of $x^2 - (k+2)x + (k+5) = 0$ are real | 4 |
| B. | <p>i) A normal to the curve $y = 3x^2 - 5x + 2$ has a gradient of -1</p> <p>Find the coordinates of the point of intersection between the normal and the curve.</p> | 3 |
| | <p>ii) Hence find where this normal cuts the y axis.</p> | 2 |

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Question 6

A. A circle:

$$(x-h)^2 + (y-k)^2 = r^2 \quad (1)$$

$$(x-2)^2 + (y+1)^2 = 9 \quad (1)$$

OR

$$\sqrt{(x-2)^2 + (y+1)^2} = 3 \quad (1)$$

$$(x-2)^2 + (y+1)^2 = 9 \quad (1)$$

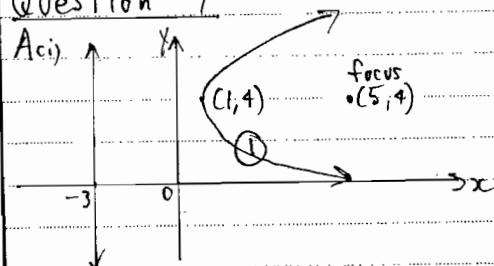
B. (i) $BA = BC$ (Sides of a square) $\angle A = \angle C$ (Angles of a square) $AE = FC$ ($ED = FD$ are subtracted from equal sides) $\therefore \triangle BAE \equiv \triangle BCF$ (SAS) (1)

(Take off one for each error or missing reason)

B. (ii) $BFC = 55$ $\Rightarrow FBC = 35$ (angle sum of a $\triangle$) (1) $\angle ABE = \angle FBC$ (corresponding angles in congruent triangles) (1) $\therefore \angle EBF = 90 - 2 \times 35$ (angle sum of a right angle)
 $= 20^\circ$ (1)

Question 7

A. (i)

(ii) $a = 4$ units (1)Focus is $(5, 4)$ (1)

(iii) Standard form:

$$(y-k)^2 = 4a(x-h) \quad (1)$$

$$(y-4)^2 = 16(x-1) \quad (1)$$

$$B. 2x^2 + 3x - 5 \equiv a(x+1)^2 + b(x+1) + c$$

$$= ax^2 + 2ax + a + bx + b + c$$

$$= ax^2 + (2a+b)x + (a+b+c)$$

$$\therefore a=2, \quad 2a+b=3, \quad a+b+c=-5$$

$$(1) \quad 2 \times 2 + b = 3 \quad 2 + 1 + c = -5$$

$$b = -1 \quad c = -6$$

(1)

(1)

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Question 8

$$A. x^2 - (k+2)x + (k+5) = 0$$

Real roots if $\Delta \geq 0$ (1)

$$\text{ie. } (k+2)^2 - 4 \times 1 \times (k+5) \geq 0 \quad (1)$$

$$k^2 + 4k + 4 - 4k - 20 \geq 0$$

$$k^2 - 16 \geq 0 \quad (1)$$

$$(k-4)(k+4) \geq 0$$

$$k \geq 4 \text{ or } k \leq -4 \quad (1)$$

$$B. y = 3x^2 - 5x + 2$$

$$\frac{dy}{dx} = 6x - 5 \quad (1)$$

Normal gradient -1 $\Rightarrow$ Tangent gradient 1

$$\Rightarrow 1 = 6x - 5 \quad (1)$$

At $x = 1$ and $y = 0$ ie. $(1, 0)$ is pt. of intersection (1)

$$m = -1, \quad (1, 0)$$

$$y - 0 = -1(x - 1) \quad (1)$$

$$y = -x + 1$$

cuts y axis when $x = 0$

$$\text{ie. at } y = 1 \quad (1)$$

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Solutions to 2005 Mathematics Assessment Task 1

Question 1

$$\begin{aligned} \text{ci) } 6x^2 + 1 & \text{ ①} & \text{cii) } 4(5x^2 - 3)^3 \times 10x & \text{ ①} & \text{ciii) } \frac{1}{2} \left(\frac{3x}{2x+5} \right) \\ & & = 40x(5x^2 - 3)^3 & \text{ ①} & = \frac{(2x+5)^3 - 3x(2)}{(2x+5)^2} \text{ ①} \\ & & = \frac{15}{(2x+5)^2} & \text{ ①} \end{aligned}$$

$$\begin{aligned} \text{civ) } (x-3)(3x^2+5) + x^3 + 5x - 6 & \text{ ①} \\ 3x^3 + 5x - 9x^2 - 15 + x^3 + 5x - 6 & \\ 4x^3 - 9x^2 + 10x - 21 & \text{ ①} \end{aligned}$$

Question 2

$$\begin{aligned} \text{A ci) } (2, -8) \text{ must satisfy} & \text{ cii) } \frac{dy}{dx} = 9x^2 - 16x \\ y = 3x^3 - 8x^2 & \text{ At } x=2, \text{ m of tangent:} \\ \text{ie: } 8 = 3 \times 2^3 - 8 \times 2^2 & 9 \times 2^2 - 16 \times 2 = 4 \text{ ①} \\ -8 = 24 - 32 & \text{ ①} \\ -8 = -8 \checkmark & \therefore y - -8 = 4(x-2) \text{ ①} \\ & 4x - y - 16 = 0 \text{ or } y = 4x \text{ ①} \end{aligned}$$

$$\text{B. PA} = \text{PB}$$

$$\begin{aligned} \sqrt{(x-2)^2 + (y-1)^2} &= \sqrt{(x-4)^2 + (y-3)^2} \text{ ①} \\ x^2 + 4x + 4 + y^2 - 2y + 1 &= x^2 - 8x + 16 + y^2 + 6y + 9 \text{ ①} \\ 12x - 8y - 20 &= 0 \\ 3x - 2y - 5 &= 0 \text{ ①} \end{aligned}$$

Question 3

$$\text{ci) Equiangular ①}$$

$$\begin{aligned} \text{cii) } \frac{x}{3.5} &= \frac{4.2}{4.2+2.8} \text{ and } \frac{y+3}{3} = \frac{4.2+2.8}{4.2} \\ x &= 2.1 \text{ ①} \text{ and } 4.2(y+3) = 21 \\ y &= 2 \text{ ①} \end{aligned}$$

Ratio of intercepts theorem OR

Corresponding sides of similar triangles make the same ratio ①

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$$\begin{aligned} \text{B ci) } \sqrt{2x-4} &= (2x-4)^{\frac{1}{2}} & \text{cii) } \frac{d}{dx} [x^2(2x-4)^{\frac{1}{2}}] \\ \therefore a &= \frac{1}{2} \text{ ①} & = x^2 \times \frac{1}{2}(2x-4)^{-\frac{1}{2}} \times 2 + 2x \\ & & = \frac{x^2}{\sqrt{2x-4}} + 2x\sqrt{2x-4} \end{aligned}$$

Question 4

$$\begin{aligned} \text{A. ci) } \alpha + \beta &= -\frac{b}{a} & \text{cii) } \alpha\beta &= \frac{c}{a} & \text{ciii) } \frac{1}{\alpha} + \frac{1}{\beta} & \text{civ) } \alpha - \beta \\ &= -3 \text{ ①} & = -5 \text{ ①} & = \frac{\alpha + \beta}{\alpha\beta} & = \frac{-3}{-5} = \frac{3}{5} \text{ ①} \end{aligned}$$

$$\text{B. } 3^{2x} - 10 \times 3^x + 9 = 0$$

$$\text{Let } v = 3^x$$

$$\begin{aligned} \therefore v^2 - 10v + 9 &= 0 \text{ ①} \\ (v-1)(v-9) &= 0 \\ v=1 &\text{ or } v=9 \text{ ①} \\ 3^x=1 &\text{ or } 3^x=9 \\ x=0 &\text{ or } 2 \text{ ①} \end{aligned}$$

Question 5

$$\begin{aligned} \text{A. ci) } y &= x^2 - 6x - 7 \\ (0, -7) &\text{ y intercept} \\ 0 &= x^2 - 6x - 7 \\ 0 &= (x+1)(x-7) \\ \Rightarrow (-1, 0) &\text{ and } (7, 0) \end{aligned}$$

$$\begin{aligned} \text{cii) } x^2 - 6x - 7 &> 0 \\ x &< -1, x > 7 \end{aligned}$$

Minimum value

B. Answer D

